AutoDD: Automated Drift Detector
Supplementary Material

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Additional Results

This section presents additional experimental results using Hoeffding Adaptive Tree (HAT) as the base classifier, demonstrating that AutoDD's performance generalizes across different base classifiers.

Synthetic Datasets: Table 1 presents results on synthetic benchmarks with known drift points using HAT as the base classifier. AutoDD achieves the highest mean accuracy and F1-score, outperforming all state-of-the-art detectors. AutoDD excels on abrupt drift scenarios where rapid detector switching is beneficial. The performance gains with HAT are even more pronounced than with GaussianNB, suggesting that AutoDD's adaptive detector selection provides greater benefits for more complex base classifiers.

Table 1: Results on synthetic datasets with HAT. Best results in **bold**.

	SEA		**HYP**		**AGR**		**RBF**		**RTG**		**Mean**	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
ADWIN	88.1	0.64	85.5	0.06	74.6	0.55	73.3	0.07	68.3	0.11	77.9	0.29
DDM	88.0	0.10	89.0	0.67	74.7	0.52	74.6	0.69	68.1	0.13	78.9	0.42
EDDM	88.0	0.46	88.6	0.68	69.2	0.29	71.8	0.42	68.8	0.47	77.3	0.46
HDDM-A	88.3	0.64	86.8	0.06	75.1	0.63	73.8	0.07	69.1	0.11	78.6	0.30
HDDM-W	88.4	0.65	89.5	0.72	75.9	0.67	73.9	0.53	69.7	0.58	79.5	0.63
Page-Hinkley	88.2	0.65	86.0	0.06	74.8	0.52	73.5	0.07	68.5	0.11	78.2	0.28
KSWIN	88.0	0.42	88.3	0.66	74.5	0.53	73.7	0.63	68.9	0.43	78.7	0.53
MDDM	89.1	0.67	90.3	0.75	75.5	0.75	73.8	0.60	69.8	0.61	78.4	0.68
ADDM	89.8	0.86	91.9	0.77	76.6	0.72	75.0	0.69	71.3	0.77	79.3	0.76
PUDD	89.8	0.83	92.6	0.83	77.1	0.76	76.4	0.74	71.9	0.74	79.7	0.78
Meta-ADD	90.9	0.91	92.5	0.92	77.1	0.81	**84.8**	0.74	**80.4**	0.79	81.4	0.83
AutoDD	**96.7**	**0.93**	**98.4**	**0.93**	**82.4**	**0.90**	75.8	**0.84**	70.9	**0.88**	**84.0**	**0.87**

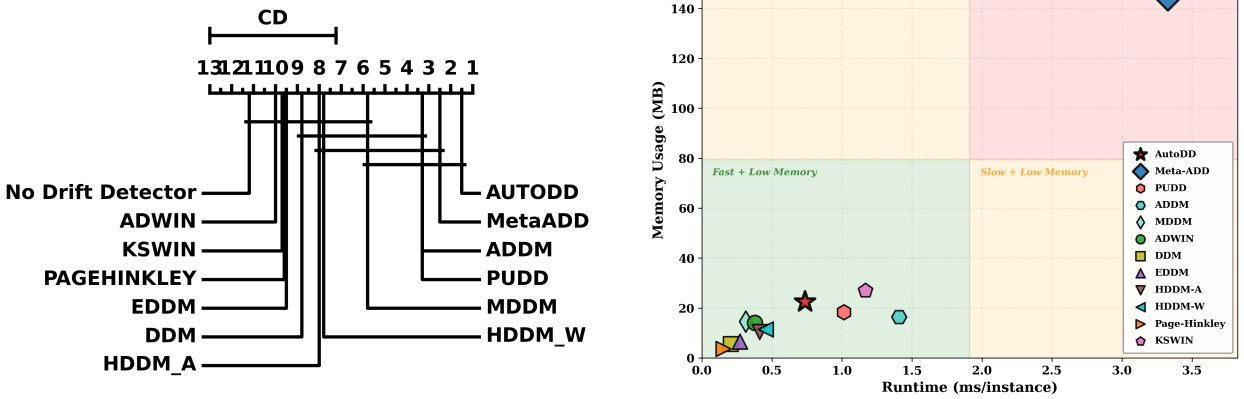
Real-world Datasets: Table 2 shows results on real-world benchmarks where drift patterns are unknown. AutoDD achieves the highest mean accuracy, outperforming all state-of-the-art detectors. AutoDD consistently ranks first across all five datasets. The larger performance gains on real-world data with HAT suggest that AutoDD's adaptive selection is particularly valuable when drift patterns are complex and unknown.

Statistical Significance: Figure 2 shows the Critical Difference diagram based on Friedman test with Nemenyi post-hoc analysis across all methods using HAT as the base classifier. AutoDD achieves the best average rank, significantly outperforming all fixed detectors and ranking substantially better than state-of-the-art methods.

Computational Efficiency: Figure 3 compares runtime and memory across methods using HAT as the base classifier. AutoDD maintains competitive efficiency, being significantly faster than Meta-ADD while achieving higher accuracy and maintaining classical detector-level computational overhead while achieving state-of-the-art performance.

Table 2: Results on real-world datasets with HAT. DN = number of detected drifts. Best results in **bold**.

	Elec		Weather		Spam		Airlines		Poker		Mean	
	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN
ADWIN	82.4	22	73.1	9	94.6	2	66.3	76	57.7	3	74.8	22.4
DDM	85.5	22	73.5	7	93.9	2	65.2	20	58.0	3	75.2	10.8
EDDM	85.2	22	77.1	9	94.8	2	64.6	182	57.8	4	75.9	43.8
HDDM-A	86.3	22	75.8	9	94.7	2	66.1	114	58.5	6	76.3	30.6
HDDM-W	85.1	22	76.0	9	94.5	1	64.9	269	56.8	404	75.5	141.0
Page-Hinkley	81.7	16	72.8	6	94.2	1	66.5	64	57.5	17	74.5	20.8
KSWIN	84.3	22	71.2	9	94.9	2	65.4	269	57.0	266	74.6	113.6
MDDM	86.5	22	78.0	9	94.6	2	66.0	110	60.0	190	81.5	66.6
ADDM	87.3	22	79.2	9	95.8	2	66.9	74	59.6	177	82.6	56.8
PUDD	87.5	22	78.6	9	95.6	2	67.7	117	59.7	81	82.9	46.2
Meta-ADD	87.9	22	78.2	9	95.9	2	66.5	139	59.6	182	83.6	70.8
AutoDD	**93.8**	22	**85.1**	9	**99.0**	2	**72.6**	118	**65.1**	188	**87.1**	67.8



(a) Critical Difference diagram

(b) Runtime vs. memory trade-off

Figure 1: Statistical significance analysis and computational efficiency with HAT base classifier. (a) Critical Difference diagram with connecting bars indicating statistically equivalent methods ($\alpha = 0.05$). (b) Runtime vs. memory trade-off showing AutoDD achieves the best accuracy while maintaining reasonable computational overhead.